

Information Gathering: Interactive Methods

LEARNING OBJECTIVES

Once you have mastered the material in this chapter, you will be able to:

1. Recognize the value of using interactive methods for information gathering.
2. Construct interview questions to elicit human information requirements and structure them in a way that is meaningful to users.
3. Understand the purpose of listening to user stories and why they are useful in systems analysis.
4. Understand the concept of JAD and when to use it.
5. Write effective questions to survey users about their work.
6. Design and administer effective questionnaires.

You can use three key interactive methods to elicit human information requirements from organization members. These three methods are interviewing and eliciting stories, joint application design (JAD), and surveying people through questionnaires. Although different in their implementation, these methods also have a great deal in common. The basis of their shared properties is talking with and listening to people who can tell stories about their interactions with technology.

Each of the three interactive methods for information gathering possesses its own established process for you to follow in interacting with users. If followed, these systematic approaches will help ensure proper design and implementation of interviews, JAD workshops, and questionnaires, as well as support insightful analysis of the resulting data. Unobtrusive methods such as sampling, investigation, and observing a decision maker's behavior and physical environment do not require the same degree of interactivity between analysts and users. These methods are covered in an upcoming chapter. By using interactive methods with unobtrusive methods, you will achieve a more complete portrait of the organization's information requirements.

Interviewing

Before you interview someone, you must, in effect, interview yourself. You need to know your biases and how they will affect your perceptions. Your education, intellect, upbringing, emotions, and ethical framework all serve as powerful filters for what you will be hearing in your interviews.

Thoroughly think through an interview before you go. Visualize why you are going, what you will ask, and what will make it a successful interview in your eyes. You must anticipate how to make the interview fulfilling for the individual you interview as well.

An information-gathering interview is a directed conversation with a specific purpose that uses a question-and-answer format. In the interview, you want to get the opinions of the interviewee and his or her feelings about the current state of the system, organizational and personal goals, and informal procedures for interacting with information technologies.

Above all, seek the opinions of the person you are interviewing. Opinions may be more important and more revealing than facts. For example, imagine asking the owner of a traditional store who has recently added an online store how many customer refunds she typically gives for Web transactions each week. She replies, “About 20 to 25 a week.” When you monitor the transactions and discover that the average is only 10.5 per week, you might conclude that the owner is overstating the facts and the problem.

Imagine instead that you ask the owner what her major concerns are and that she replies, “In my opinion, customer returns of goods purchased over the Web are way too high.” By seeking opinions rather than facts, you discover a key problem that the owner wants addressed.

In addition to opinions, you should try to capture the feelings of the interviewee. Remember that the interviewee knows the organization better than you do. You can understand the organization’s culture more fully by listening to the feelings of the respondent.

Goals are important information that can be gleaned from interviewing. Facts that you obtain from hard data may explain past performance, but goals project the organization’s future. Try to find out as many of the organization’s goals as possible from interviewing. You may not be able to determine goals through any other data-gathering methods.

The interview is also a valuable opportunity to explore key human–computer interaction (HCI) concerns, including the ergonomic aspects, system usability, how pleasing and enjoyable the system is, and how useful it is in supporting individual tasks.

In the interview, you are setting up a relationship with someone who is probably a stranger to you. You need to build trust and understanding quickly, but at the same time you must maintain control of the interview. You also need to sell the system by providing needed information to your interviewee. Do so by planning for the interview before you go so that conducting it is second nature to you. Later in the chapter, we discuss joint application design (JAD) (pronounced “jad,” rhymes with *add*), which can serve as an alternative to one-on-one interviewing in certain situations.

Five Steps in Interview Preparation

The five major steps in interview preparation are shown in Figure 4.1. These steps include a range of activities, from gathering basic background material to deciding who to interview.

READING BACKGROUND MATERIAL It’s important to read and understand as much background information about the interviewees and their organization as possible. This material often can be obtained on the corporate website, from a current annual report, a corporate newsletter, or any publications sent out to explain the organization to the public. Check the Internet for any corporate information, such as information from Standard & Poor’s.

FIGURE 4.1

Steps a systems analyst follows in planning an interview.

Steps in Planning the Interview	
1.	Read background material.
2.	Establish interviewing objectives.
3.	Decide whom to interview.
4.	Prepare the interviewee.
5.	Decide on question types and structure.

As you read through this material, be particularly sensitive to the language organization members use in describing themselves and their organization. What you are trying to do is build up a common vocabulary that will eventually enable you to phrase interview questions in a way that is understandable to your interviewee. Another benefit of researching your organization is to optimize the time you spend in interviews; without such preparation, you may waste time asking general background questions.

ESTABLISHING INTERVIEWING OBJECTIVES Use the background information you gathered as well as your own experience to establish interview objectives. There should be four to six key areas concerning HCI, information processing, and decision-making behavior about which you will want to ask questions. These areas include HCI concerns (the usefulness and usability of the system; how it fits physical aspects; how it suits a user's cognitive capabilities; whether it is engaging or aesthetically pleasing; and whether using the system is rewarded with desired consequences), information sources, information formats, decision-making frequency, qualities of information, and decision-making style.

DECIDING WHOM TO INTERVIEW When deciding whom to interview, include key people at all levels who will be affected by the system in some manner. Strive for a balance that addresses as many users' needs as possible. Your organization contact will also have some ideas about who should be interviewed.

PREPARING THE INTERVIEWEE Prepare the person to be interviewed by calling ahead or sending an email message and allowing the interviewee time to think about the interview. If you are doing an in-depth interview, it is permissible to email your questions ahead of time to allow the interviewee time to think over his or her responses. Because there are many objectives to fulfill in the interview (including building trust and observing the workplace) interviews should typically be conducted in person and not via email. Interviews should be kept to 45 minutes or an hour at most. No matter how much your interviewees seem to want to extend the interview beyond this limit, remember that when they spend time with you, they are not doing their work. If interviews go over an hour, it is likely that the interviewees will resent the intrusion, whether or not they articulate their resentment.

DECIDING ON QUESTION TYPES AND STRUCTURE Write questions to cover the key areas of HCI and decision making that you discovered when you ascertained interview objectives. Proper questioning techniques are the heart of interviewing. Questions have some basic forms you need to know. The two basic question types are open-ended and closed. Each question type can accomplish something a little different from the other, and each has benefits and drawbacks. You need to think about the effect each question type will have.

It is possible to structure your interview in three different patterns: a pyramid structure, a funnel structure, or a diamond structure. Each is appropriate under different conditions and serves a different function, and each one is discussed later in this chapter.

Question Types

You will use three main question types when you construct your interview: open-ended questions, closed questions, and probes. Each question type is useful in different situations and at specific times during the interview. In addition, each type of question elicits different information from the person you are interviewing.

OPEN-ENDED QUESTIONS Open-ended questions include those such as "What do you think about putting all the managers on an intranet?" "Please explain how you make a scheduling decision." and "In what ways does the system extend your capability to do tasks that would not be possible otherwise?" Consider the term *open-ended*. "Open" actually describes the interviewee's options for responding. They are open. The response can be two words or two paragraphs. Some examples of open-ended questions are found in Figure 4.2.

The benefits of using open-ended questions are numerous and include the following:

1. Putting the interviewee at ease
2. Allowing the interviewer to pick up on the interviewee's vocabulary, which reflects his or her education, values, attitudes, and beliefs

FIGURE 4.2

Open-ended interview questions allow the respondent open options for responding. These examples were selected from different interviews and are not shown in any particular order.

Open-Ended Interview Questions
• What's your opinion of the current state of business-to-business ecommerce in your firm? • What are the critical objectives of your department? • Once the data are submitted via the website, how are they processed? • Describe the monitoring process that is available online. • What are some of the common data entry errors made in this department? • What are the biggest frustrations you've experienced during the transition to ecommerce?

3. Providing richness of detail
4. Revealing avenues of further questioning that otherwise may go untapped
5. Making the interview more interesting for the interviewee
6. Allowing more spontaneity
7. Making phrasing easier for the interviewer
8. Using them in a pinch if the interviewer is caught unprepared

As you can see, there are several advantages to using open-ended questions. There are, however, also many drawbacks:

1. Asking questions that may result in too much irrelevant detail
2. Possibly losing control of the interview
3. Allowing responses that may take too much time for the amount of useful information gained
4. Potentially seeming that the interviewer is unprepared
5. Possibly giving the impression that the interviewer is on a “fishing expedition,” with no real objective for the interview

You must carefully consider the implications of using open-ended questions for interviewing.

CLOSED QUESTIONS The alternative to open-ended questions is found in the other basic question type: closed questions. Such questions are of the basic form “Is it easy to use the current system?” and “How many subordinates do you have?” The possible responses are closed to the interviewee because he or she can only reply with a finite number, such as “None,” “One,” or “Fifteen.” Some examples of closed questions can be found in Figure 4.3.

A closed question limits the responses available to the interviewee. You may be familiar with closed questions through multiple-choice exams in college. You are given a question and five responses, but you are not allowed to write down your own response and get credit for correctly answering the question.

A special kind of closed question is the bipolar question. This type of question limits the interviewee even further by allowing only a choice on either pole, such as yes or no, true or false, or agree or disagree. Examples of bipolar questions can be found in Figure 4.4.

The benefits of using closed questions of either type include the following:

1. Saving time
2. Easily comparing interviews

FIGURE 4.3

Closed interview questions limit the respondent's options. These examples were selected from different interviews and are not shown in any particular order.

Closed Interview Questions
• How many times a week is the project repository updated? • On average, how many calls does the call center receive monthly? • Which of the following sources of information is most valuable to you? ◦ Completed customer complaint forms ◦ Email complaints from consumers who visit the website ◦ Face-to-face interaction with customers ◦ Returned merchandise • List your top two priorities for improving the technology infrastructure. • Who receives this input?

Bipolar Interview Questions

- Do you use the Web to provide information to vendors?
- Do you agree or disagree that ecommerce on the Web lacks security?
- Do you want to receive a printout of your account status every month?
- Does your website maintain a FAQ page for employees with payroll questions?
- Is this form complete?

FIGURE 4.4

Bipolar interview questions are a special kind of closed question. These examples were selected from different interviews and are not shown in any particular order.

3. Getting to the point
4. Keeping control over the interview
5. Covering lots of ground quickly
6. Getting to relevant data

The drawbacks of using closed questions are substantial, however. They include the following:

1. Boring for the interviewee
2. Failing to obtain rich detail (because the interviewer supplies the frame of reference for the interviewee)
3. Failing to address the main ideas for the preceding reason
4. Failing to build rapport between interviewer and interviewee

Thus, as an interviewer, you must think carefully about the question types you will use.

Both open-ended and closed questions have advantages and drawbacks, as shown in Figure 4.5. Notice that choosing one question type over the other involves a trade-off; although an open-ended question affords breadth and depth of reply, responses to open-ended questions are difficult to analyze.

PROBES A third type of question is the probe, or follow-up. The strongest probe is the simplest—the question “Why?” Other probes are “Please provide an example of a time you did not find the system trustworthy,” and “Please elaborate on that for me.” Some examples of probing questions can be found in Figure 4.6. The purpose of a probe is to go beyond the initial answer to get more meaning, to clarify, and to draw out and expand on the interviewee’s point. Probes may be either open-ended or closed questions.

It is essential to probe. Most beginning interviewers are reticent about probing and consequently accept superficial answers. They are usually grateful that employees have granted interviews and feel somewhat obligated to accept unqualified statements politely.

Arranging Questions in a Logical Sequence

Just as there are two generally recognized ways of reasoning—inductive and deductive—there are two similar ways of organizing your interviews. A third way combines both inductive and deductive patterns.

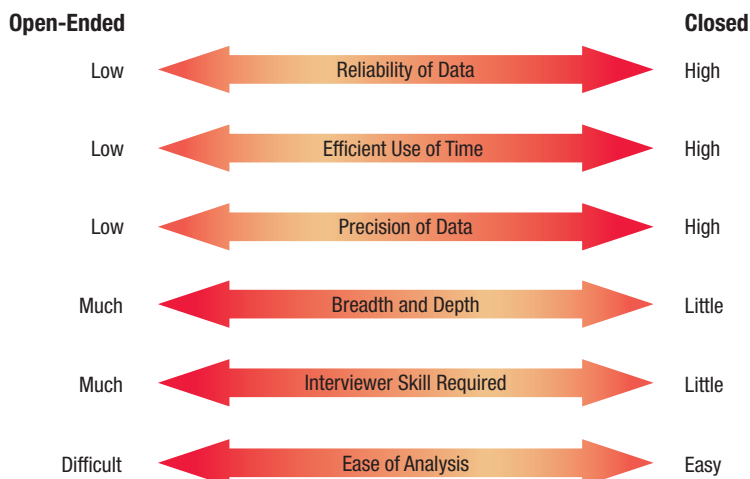


FIGURE 4.5

Attributes of open-ended and closed questions.



CONSULTING OPPORTUNITY 4.1

Strengthening Your Question Types

Strongbodies, a large local chain of sports clubs, has experienced phenomenal growth in the past five years. Management would like to refine its decision-making process for purchasing new body-building equipment by developing a dashboard. Currently, managers listen to customers, attend trade shows, look at advertisements, and put in requests for new equipment purchases based on their subjective perceptions. These are then approved or denied by Harry Mussels.

Harry is the first person you will interview. He is a 37-year-old division manager who runs five area clubs. He travels all over the city to their widespread locations. He keeps an office at the East location, although he is there less than a quarter of the time.

In addition, when Harry is present at a club, he is busy answering business-related phone calls, solving on-the-spot

problems presented by managers, and interacting with club members. His time is short, and to compensate for that he has become an extremely well-organized, efficient divisional manager. He cannot grant you a lot of interview time. However, his input is important, and he feels he would be the main beneficiary of the proposed system.

What type of interview question might be most suitable for your interview with Harry? Why is this type most appropriate? How will your choice of question type affect the amount of time you spend in preparation for interviewing Harry? Write 5 to 10 questions of this type. What other techniques might you use to supplement information unavailable through that type of question? Write a paragraph to explain.

USING A PYRAMID STRUCTURE Inductive organization of interview questions can be visualized as having a pyramid shape. Using this form, the interviewer begins with very detailed, often closed, questions. The interviewer then expands the topics by allowing open-ended questions and more generalized responses, as shown in Figure 4.7.

You should use a pyramid structure if you believe your interviewee needs to warm up to the topic. Using a pyramid structure for question sequencing is also useful when you want an ending determination about the topic. Such is the case in the final question in Figure 4.7, “In general, how do you feel about the security of data versus the importance of Internet access?”

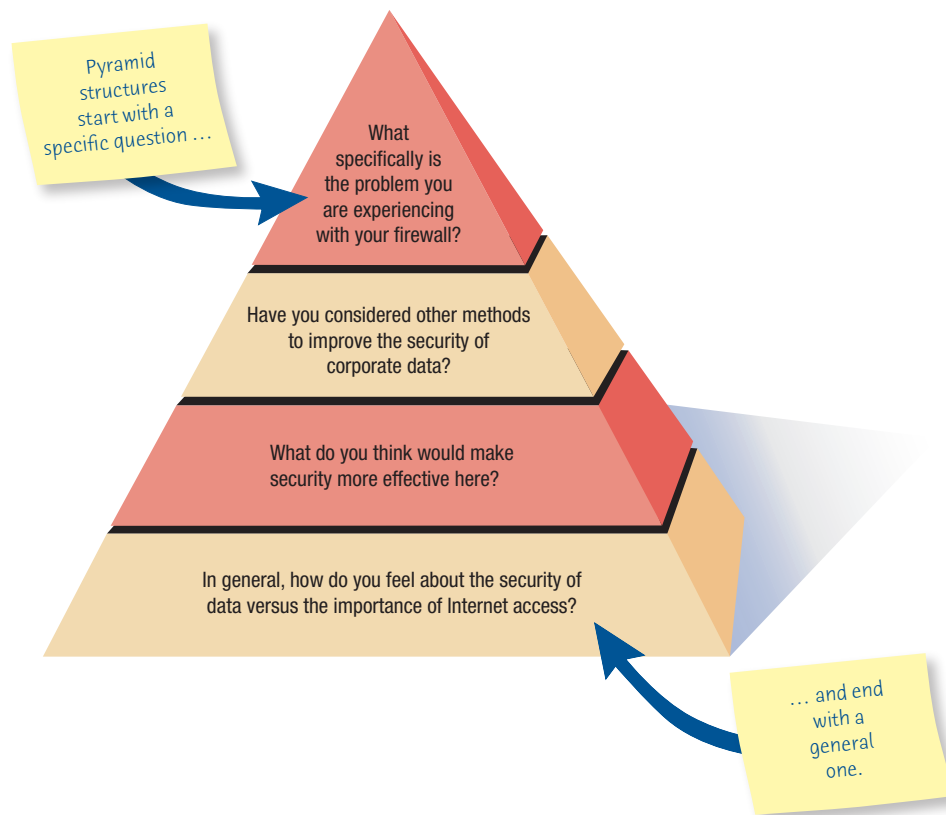
USING A FUNNEL STRUCTURE In the second kind of structure, the interviewer takes a deductive approach by beginning with generalized, open-ended questions and then narrowing the possible responses by using closed questions. This interview structure can be thought of as funnel shaped, as depicted in Figure 4.8. Using the funnel structure method provides an easy, nonthreatening way to begin an interview. A funnel-shaped question sequence is also useful when the interviewee feels emotional about the topic and needs freedom to express those emotions.

USING A DIAMOND-SHAPED STRUCTURE Often a combination of the pyramid and funnel structures, resulting in a diamond-shaped interview structure, is best. This structure entails beginning in a very specific way, then examining general issues, and finally coming to a very specific conclusion, as shown in Figure 4.9.

FIGURE 4.6

Probes allow a systems analyst to follow up on questions to get more detailed responses. These examples were selected from different interviews and are not shown in any particular order.

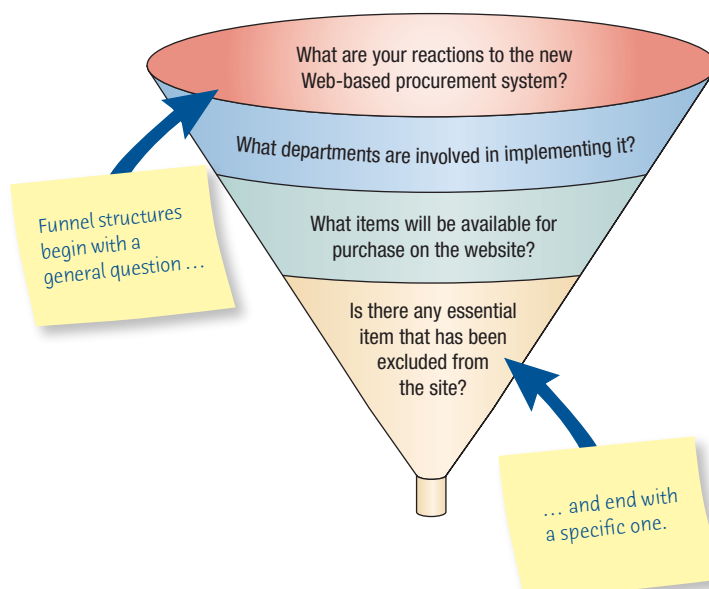
Probes
• Why? • Give an example of how ecommerce has been integrated into your business processes. • Please give an illustration of the security problems you are experiencing with your online bill payment system. • You mentioned both an intranet and an extranet solution. Please give an example of how you think each differs. • What makes you feel that way? • Tell me step by step what happens after a customer clicks the “Submit” button on the Web registration form.

**FIGURE 4.7**

The pyramid structure for interviewing goes from specific to general questions.

The interviewer begins with easy, closed questions that provide a warm-up to the interview process. In the middle of the interview, the interviewee is asked for opinions on broad topics that obviously have no “right” answer. The interviewer then narrows the questions again to get specific questions answered, thus providing closure for both the interviewee and the interviewer. The diamond structure combines the strengths of the other two approaches but has the disadvantage of taking longer than either other structure.

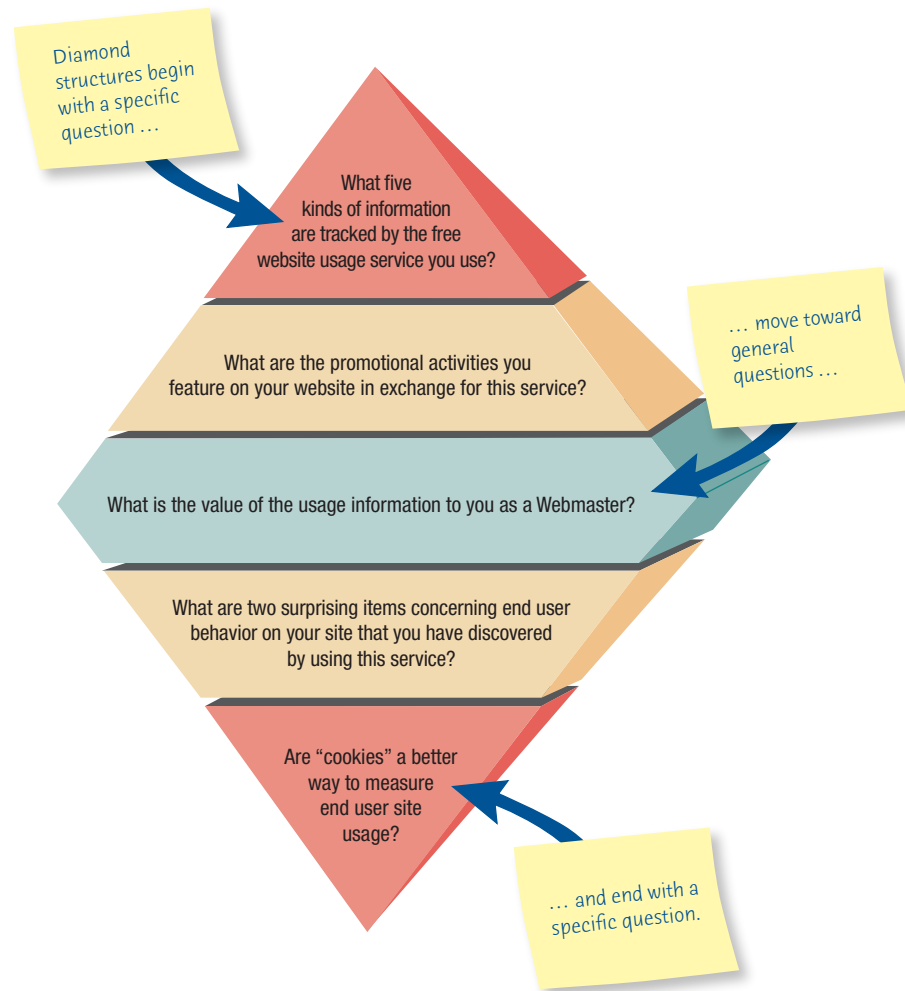
The end of the interview is a natural place to ask one key question: “Is there anything we haven’t touched on that you feel is important for me to know?” Considered a formula question by the interviewee most of the time, the response will often be “No.” You are interested

**FIGURE 4.8**

The funnel structure for interviewing begins with broad questions and then funnels to specific questions.

FIGURE 4.9

The diamond-shaped structure for interviewing combines the pyramid and funnel structures.



in the other times, when this question opens the proverbial floodgates and much new data are presented.

As you conclude the interview, summarize and provide feedback on your overall impressions. Inform the interviewee about the subsequent steps to take and what you and other team members will do next. Ask the interviewee with whom you should talk next. Set up future appointment times for follow-up interviews, thank the interviewee for his or her time, and shake hands.

Writing the Interview Report

Although the interview itself is complete, your work on the interview data is just beginning. You need to capture the essence of the interview through a written report. It is imperative that you write the interview report as soon as possible after the interview. This step is another way you can ensure quality of interview data. The longer you wait to write up your interview, the more suspect the quality of your data becomes.

After this initial summary, go into more detail, noting main points of the interview and your own opinions. Review the interview report with the respondent at a follow-up meeting. This step helps clarify the meaning the interviewee had in mind and lets the interviewee know that you are interested enough to take the time to understand his or her point of view and perceptions.

Listening to Stories

Stories originate in the workplace, and many are shared with, and repeated by, coworkers. Organizational stories, like myths and fables, are used to relay some kind of information. This shared information is usually considered important enough to build an entire story around it, and when a



CONSULTING OPPORTUNITY 4.2

Skimming the Surface

You are about to leave SureCheck Dairy after a preliminary tour when another member of your systems analysis team calls you at the dairy to say he cannot make his interview appointment with the plant manager because of illness. The plant manager is extremely busy, and you want to keep his enthusiasm for the project going by doing things as scheduled. You also realize that without the initial interview data, the rest of your data gathering will be slowed. Although you have no interview questions prepared, you make the decision to go ahead and interview the plant manager on the spot.

You have learned that sales managers at SureCheck are all interested in processing their own data and producing their own reports on quantities and kinds of dairy products sold so that they

can use that information to better control production of the company's large product line (it includes whole, skim, 2 percent, and 1 percent milk, half-and-half, cottage cheese, yogurt, and frozen novelties). Sales managers are currently sending their sales figures via the Web to corporate headquarters, 600 miles away, and even the production of online reports seems slow. You will base your ad-libbed questions on what you have just found out on the tour.

In the few minutes before your interview begins, decide on a structure for it: funnel, pyramid, or diamond. In a paragraph, justify why you would proceed with the interview structure you have chosen based on the unusual context of this interview. Write a series of questions and organize them in the structure you have chosen.

story is told and retold, even handed down over time, the story takes on a mythic quality. Isolated stories are welcome when you are looking for facts, and enduring stories capture all aspects of the organization. Enduring stories are the ones a systems analyst should be seeking.

Although structured interviews have many advantages, including the ability to ask many people the same set of questions, they may not produce the rich detail you need to design your system.

Suppose the developer is faced with the task of analyzing and redesigning a system. The developer can learn from experiences the organization encountered in their previous design project. It would be helpful to identify people who had lived through that experience. They may have valuable insights that will help the developer with the new project.

It is difficult for the developer to generate a list of interview questions because the problems encountered, the obstacles cleared, and the learning that took place are unknown to the developer. By encouraging stories, however, these experiences may be brought out, maybe not at first, but as the story is being told. Patience is a key value of the listener.

The developer may be able to understand the behavior and the values of organization members by listening. By listening carefully to organization members, the developer can get a much better understanding of the strategic objectives, potential obstacles, decisions, and lessons learned.

It might be apparent that listening to stories is not efficient. It takes considerably more time than asking interview questions. However, listening to stories may be more rewarding. Stories are more easily remembered than interview responses, so in some ways that makes up for the fact that it takes time to listen.

Stories Are Made Up of Elements

Storytellers often make themselves the hero of the story. This is not unusual and it is not a drawback. When storytellers put themselves into the center of the story, the struggles they encounter are their own battles. They become real to the listener who can empathize with the storytellers.

An example of a story is shown in Figure 4.10. All stories have these elements:

1. The call to adventure
2. The quest
3. The struggle
4. The transformation
5. The resolution
6. The moral
7. The epilogue

FIGURE 4.10

Story elements and corresponding examples. Quotation from “Open source software opens door to Web design career” by Mark Thomas (2016).

Story element	Example from “Open source software opens door to Web design career” by Mark Thomas (2016)
The call to adventure	“I was fascinated and envious—I wanted to be able to create those exciting images with their grungy textures, characters lifted from games, dramatic lighting, and emphatic quotes. Other people didn’t seem to find it difficult to finish them, but I had no idea where to even start.”
The quest	“I was going to have to learn this discipline properly. Such things are not achieved overnight, of course, and so began a long quest to find, practice, and learn from every design resource I could find.”
The struggle	“Researching the topic, I realized that I had used the wrong tool for the job. Exasperated and in no small way desperate, I began looking for a package I could turn to rescue the situation. Time was pressing and, even if I could find an answer in time, as the now self-proclaimed designer in the team I would have to learn it double-quick.”
The transformation	“Taking stock for a moment, I began to wonder if it was possible to begin a business operation based on the open source software I was now developing in confidence and ability with. It became an ambition.”
The resolution	“In 2009, I designed and built a website for the bakery that went on to supply our wedding cake, saving us money and introducing new customers to their business, too. In 2010, I created all the artwork for our wedding stationery. Later the same year, I registered my UK company and began providing Web design services to local companies who needed designs for their websites, which I then started to code.”
The moral	“To me, open source software means opportunity. It means education, freedom, commitment, collaboration and challenges. I went from knowing nothing whatsoever about design and coding to starting my own Web design business and capturing local clients who were happy with the things I’ve made for them and unconcerned by the tools I’d used to do it.”
The epilogue	“And even though my day job prevents me from publishing anywhere near the number of blog articles I want to (for which I apologize to my readers—you are not forgotten), it still manages nearly 3,000 visitors a month, and I regularly receive requests for tutorials from around the world, which I will get around to at some point.”

A typical story starts with a reason an opportunity was noticed or a problem needed addressing. This is the call to adventure, and there is never a simple solution. No one tells a story such as “It was drafty last night, so I got up and shut the window.” That is a statement not a story. Stories have twist and turns, advances and setbacks, learning and decision making.

The quest in a story implies that there is something worth finding or achieving. In explaining the quest, the storyteller enumerates the organization’s objectives. It is worthwhile to undergo hardship if there is a worthwhile goal.

Next comes the struggle. Listen to hear who the storyteller names as villains. Is it government regulation, external competition, or other members of the organization? It helps to understand what obstacles may exist now.

If a solution existed, the problem would be solved and the story would end here, but simply pushing a metaphorical boulder out of the way does not typically solve problems. Usually, something within the organization, or even the hero, has to change. Problems are solved by process or behavior changes rather than by product changes, and this is revealed in this part of the story. Storytellers also reveal their own weaknesses. In this revelatory approach, the storyteller describes overcoming this weakness, even undergoing a phenomenal transformation. Once the hero undergoes the transformation, it becomes possible to solve the problem.

The resolution comes when the storyteller tells whether or not the quest succeeded. It is possible that the storyteller may be telling you how a technology adoption failed, but you are more likely to hear a positive story.

Usually a moral is important to the story. The reason for the moral is to tell others how to behave and, furthermore, what values they should possess. It is a statement of the lessons learned.

Finally, the story may end with an epilogue. A happy resolution when the problem was first solved may be changed with the passage of time. Competition, customer preferences, improved hardware, and social, environmental, or legal issues that have occurred since the successful implementation may be creating new obstacles for the organization. Perhaps that is why the developer has been called in to change the system again.

Reasons for Telling Stories

The rich information an analyst gets from listening carefully to the stories is in itself valuable, but the analyst shouldn't stop there. The information gleaned from the stories will make more sense and be more valuable if seen in context. Why did the storyteller tell this story? In childhood, we are told stories and fables so that we adopt good values. For example, the fables about "The Grasshopper and the Ant" or "The Hare and the Tortoise" are commonly told in English-speaking cultures. The former teaches us the importance of good work habits and the latter emphasizes what happens if we are overconfident. Do business stories have the same purpose?

Business stories are a little more complex. We can break down stories into four important main types:

1. Experiential stories
2. Explanatory stories
3. Validating stories
4. Prescriptive stories

Experiential stories describe what life is like in the organization. Explanatory stories go a little further, suggesting that there is an explanation for behavior or a reason for the decisions made within the organization. Validating stories are used to convince people that the organization made the correct decision, or they are used to express and maintain the existing values of the organization. Prescriptive stories are the organizational equivalent of childhood fables. These stories suggest how everyone in the organization *should* behave. It is helpful for the developer to know why a storyteller is telling his or her story.

Systems analysts can use storytelling as a complement to other information-gathering methods such as interviewing, JAD, and surveys. You need to engage organization participants by reacting to stories, matching one story to another by recounting it to other participants, and even collaborating with the participant to reframe and understand organization stories. It is a good way to deeply understand some of the problems associated with information systems use, systems development, systems adoption, and designing for your intended audience.

We have observed that many of the stories systems analysts have reported on in the past have been mere fragments of stories. The intent or purpose of the story is not understood, and therefore the analyst is not able to fully understand the importance of the story. The systems analyst needs to listen to the story as a whole to truly understand its content and purpose.

Joint Application Design

No matter how adept you become as an interviewer, you will inevitably experience situations in which one-on-one interviews do not seem to be as useful as you would like. Personal interviews are time consuming and subject to error, and their data are prone to misinterpretation. IBM developed an alternative approach to interviewing users one by one, called joint application design (JAD). The motivations for using JAD are to cut the time (and hence the cost) required for personal interviews, to improve the quality of the results of information requirements assessment, and to create more user identification with new information systems as a result of the participative processes.

Although JAD can be substituted for personal interviews at any appropriate juncture during the systems development life cycle (SDLC), it has usually been employed as a technique that allows you, as a systems analyst, to accomplish requirements analysis and to design the user interface jointly with users in a group setting. The many intricacies of this approach can be learned only in a paid seminar demonstrating proprietary methods. We can, however, convey enough information about JAD here to make you aware of some of its benefits and drawbacks in comparison with one-on-one interviews.

Conditions That Support the Use of JAD

The following list of conditions will help you decide when the use of JAD may be fruitful. Consider using joint application design when:

1. User groups are restless and want something new, not a standard solution to a typical problem.
2. The organization culture supports joint problem-solving behaviors among multiple levels of employees.
3. Analysts forecast that the number of ideas generated via one-on-one interviews will not be as plentiful as the number of ideas possible from an extended group exercise.
4. Organization workflow permits the absence of key personnel during a two- to four-day block of time.

Who Is Involved?

Joint application design sessions include a variety of participants—analysts, users, executives, and so on—who contribute different backgrounds and skills to the sessions. Your primary concern here is that all project team members are committed to the JAD approach and become involved. Choose an executive sponsor, a senior person who will introduce and conclude the JAD session. Preferably, select an executive from the user group who has some sort of authority over the information systems (IS) people working on the project. This person will be an important, visible symbol of organizational commitment to the systems project.

At least one IS analyst should be present, but the analyst usually takes a passive role, unlike traditional interviewing in which the analyst controls the interaction. As the project analyst, you should be present during JAD to listen to what users say and what they require. In addition, you will want to give an expert opinion about any disproportionate costs of solutions proposed during the JAD session itself. Without this kind of immediate feedback, unrealistic solutions with excessive costs may creep into the proposal and prove costly to discourage later on.

From eight to a dozen users, of any rank, can be chosen to participate in JAD sessions. Try to select users who can articulate what information they need to perform their jobs as well as what they desire in a new or improved computer system.

The session leader should not be an expert in systems analysis and design but rather someone who has excellent communication skills to facilitate appropriate interactions. Note that you do not want to use a session leader who reports to another person in the group. To avoid this possibility, an organization may want to retain an outside management consultant to serve as session leader. The point is to get a person who can bring the group's attention to important systems issues, satisfactorily negotiate and resolve conflicts, and help group members reach a consensus.

Your JAD session should also include one or two observers who are analysts or technical experts from other functional areas to offer technical explanations and advice to the group during the sessions. In addition, one scribe from the IS department should attend the JAD sessions to formally write down everything that is done.

Where to Hold JAD Meetings

If at all possible, we recommend holding the two- to four-day sessions off-site, away from the organization, in comfortable surroundings. Some groups use executive centers or even group decision support facilities that are available at major universities. The idea is to minimize the daily distractions and responsibilities of the participants' regular work. The room itself should comfortably hold the number of people invited. Minimal presentation support equipment includes two projectors to connect to two laptops, iPads, or PCs; a whiteboard; a flip chart; and easy access to a copier. Group decision support rooms also provide networked PCs, a projection system, and software written to facilitate group interaction while minimizing unproductive group behaviors.

Schedule your JAD session when all participants can commit to attending. Do not hold the sessions unless everyone who has been invited can actually attend. This rule is critical to the success of the sessions. Ensure that every participant receives an agenda before the meeting, and consider holding an orientation meeting for a half day one week or so before the workshop so that those involved know what is expected of them. Such a premeeting allows you to move rapidly and act confidently once the actual meeting is convened.



HYPERCASE EXPERIENCE 4.1

“Well, I did warn you that things aren’t always smooth here at MRE. By now you’ve met many of our key employees and are starting to understand the ‘lay of the land.’ Who would have thought that some innocent decisions about systems, like whether to purchase a COMTEX or Shiroma, would cause such hostility? Well, live and learn, I always say. At least now you’ll know what you’re up against when you have to start recommending software and hardware!”

“It’s funny that not all questions are created equal. I myself favor asking open-ended questions, but when I have to answer them, it is not always easy.”

HYPERCASE Questions

1. Using the interview questions posed in HyperCase, give five examples of open-ended questions and five examples of closed questions. Explain why your examples are correctly classified as either open-ended or closed question types.
2. List three probing questions that are part of the Daniel Hill interview. In particular, what did you learn by following up on the questions you asked Daniel?
3. List three probing questions that are part of the Mr. Evans interview. In particular, what did you learn by following up on the questions you asked Mr. Evans?



FIGURE 4.HC1

Pointing to a question in HyperCase will reveal an answer.

Accomplishing a Structured Analysis of Project Activities

IBM recommends that the JAD sessions examine these points in the proposed systems project: planning, receiving, receipt processing/tracking, monitoring and assigning, processing, recording, sending, and evaluating. For each topic, the questions who, what, how, where, and why should be asked and answered as well. Clearly, ad hoc interactive systems such as recommender systems and other types of systems dependent on decision-maker style (including prototype systems) are not as easily analyzed with the structured approach of JAD.

As the analyst involved with the JAD sessions, you should receive the notes of the scribe and prepare a specifications document based on what happened at the meeting. Systematically present the management objectives as well as the scope and boundaries of the project. Specifics of the system, including details on screen and report layouts, also should be included.

Potential Benefits of Using JAD in Place of Traditional Interviewing

There are four major potential benefits that you, the users, and your systems analysis team should consider when you weigh the possibilities of using JAD. The first potential benefit is timesavings over traditional one-on-one interviews. Some organizations have estimated that JAD sessions have provided a 15 percent timesavings over the traditional approach.

Hand in hand with timesavings is the rapid development possible via JAD. Because user interviews are not accomplished serially over a period of weeks or months, the development can proceed much more quickly.

A third benefit to weigh is the possibility of improved ownership of the information system. Analysts are always striving to involve users in meaningful ways and to encourage users to take early ownership of the systems being designed. Due to its interactive nature and high visibility, JAD helps users become involved early in systems projects and treats their feedback seriously. Working through a JAD session eventually helps reflect user ideas in the final design.

A final benefit of participating in JAD sessions is the creative development of designs. The interactive character of JAD has a great deal in common with brainstorming techniques that generate new ideas and new combinations of ideas because of the dynamic and stimulating environment. Designs evolve through facilitated interactions rather than in relative isolation.

Potential Drawbacks of Using JAD

There are three drawbacks or pitfalls that you should weigh when making a decision on whether to do traditional one-on-one interviews or to use JAD. The first drawback is that JAD requires the commitment of a large block of time from all participants. Because JAD requires a two- to four-day commitment, it is not possible to do any other activities concurrently or to time-shift any activities, as is typically done in one-on-one interviewing.

A second pitfall occurs if preparation for the JAD sessions is inadequate in any regard or if the follow-up report and documentation of specifications is incomplete. In these instances, resulting designs could be less than satisfactory. Many variables need to come together correctly for JAD to be successful. Conversely, many things can go wrong. The success of designs resulting from JAD sessions is less predictable than that achieved through standard interviews.

Finally, the necessary organizational skills and organization culture may not be sufficiently developed to enable the concerted effort required to be productive in a JAD setting. In the end you will have to judge whether the organization is truly committed to, and prepared for, this approach.

Using Questionnaires

The use of questionnaires is an information-gathering technique that allows systems analysts to study attitudes, beliefs, behavior, and characteristics of several key people in the organization who may be affected by the current and proposed systems. Attitudes are what people in the organization say they want (in a new system, for instance); beliefs are what people think is actually true; behavior is what organization members do; and characteristics are properties of people or things.

Responses gained through questionnaires (also called surveys) using closed questions can be quantified. If you are surveying people via email or the Web, you can use software to turn electronic responses directly into data tables for analysis using a spreadsheet application or statistical software package. Responses to questionnaires using open-ended questions are analyzed and interpreted in other ways. Answers to questions on attitudes and beliefs are sensitive to the wording chosen by the systems analyst.

Through the use of questionnaires, the analyst may be seeking to quantify what was found in interviews. In addition, questionnaires may be used to determine how widespread or limited a sentiment expressed in an interview really is. Conversely, questionnaires can be used to survey a large sample of system users to sense problems or raise important issues before interviews are scheduled.

Throughout this chapter, we compare and contrast questionnaires with interviews. There are many similarities between the two techniques, and perhaps the ideal would be to use them in conjunction with each other, either following up unclear questionnaire responses with an interview or designing the questionnaire based on what is discovered in the interview. Each technique, however, has its own specific functions, and it is not always necessary or desirable to use both.



CONSULTING OPPORTUNITY 4.3

A Systems Analyst, I Presume?

“**K**now what I think of the work the last systems analyst team did? The proliferation of PDF reports created a jungle on my desk-top. To figure out the cost of raw materials to us, I have to cut my way through the overgrowth of data, hacking my path with drill-down capabilities that I have had to create for myself. I delete everything that’s irrelevant. Sometimes I rip out the excess vegetation until I see the numbers I need on my screen,” says Henry Stanley, accounting supervisor for Zenith Glass Company. As you interview him, he points unhappily to an untidy stack of old print-outs sprouting beside his desk. “Those were output from the work

of a systems team who worked here a dozen years ago. I still keep them because you never know when you’ll need them as a guide. It’s definitely survival of the fittest around here.”

Identify the overriding metaphor Henry is using to describe the reports he is receiving and the accessibility of information in them. In a paragraph, describe how this step helps you understand Henry’s attitude toward any work proposed by your systems analysis team. In a paragraph, adopt Henry’s metaphor and extend it in a more positive sense during your interview with him by writing six interview questions based on his metaphor.

Planning for the Use of Questionnaires

At first glance, using questionnaires may seem to be a quick way to gather massive amounts of data about how users assess the current system, about what problems they are experiencing with their work, and about what people expect from a new or modified system. Although it is true that you can gather a lot of information through questionnaires without spending time in face-to-face interviews, developing a useful questionnaire takes extensive planning time in its own right. When you decide to survey users via email or the Web, you face additional planning considerations concerning confidentiality, authentication of identity, and problems of multiple responses.

You must first decide what you are attempting to gain through using a survey. For instance, if you want to know what percentage of users prefers a FAQ page as a means of learning about new software packages, a questionnaire might be the right technique. If you want an in-depth analysis of a manager’s decision-making process, conducting an interview is a better choice.

Here are some guidelines to help you decide whether the use of questionnaires is appropriate. Consider using questionnaires if:

1. The people you need to question are widely dispersed (as in different branches of the same corporation).
2. A large number of people are involved in the systems project, and it is meaningful to know what proportion of a given group (for example, management) approves or disapproves of a particular feature of the proposed system.
3. You are doing an exploratory study and want to gauge overall opinion before the systems project is given any specific direction.
4. You wish to be certain that any problems with the current system are identified and addressed in follow-up interviews.

Once you have determined that you have good cause to use a questionnaire and have pinpointed the objectives to be fulfilled through its use, you can begin formulating questions.

Writing Questions

The biggest difference between the questions used for most interviews and those used on questionnaires is that interviewing permits interaction between the questions and their meanings. In an interview, the analyst has an opportunity to refine a question, define a muddy term, change the course of questioning, respond to a puzzled look, and generally control the context.

Few of these opportunities are possible on a questionnaire. Thus, for an analyst, questions must be transparently clear, the flow of the questionnaire cogent, the respondent’s questions anticipated, and the administration of the questionnaire planned in detail. (A respondent is the person who responds to or answers the questionnaire.)

The basic question types used on the questionnaire are open-ended and closed, as discussed for interviewing. Due to the constraints placed on questionnaires, some additional discussion of question types is warranted.

OPEN-ENDED QUESTIONS Recall that open-ended questions (or statements) are those that leave all possible response options open to the respondent. For example, open-ended questions on a questionnaire might read, “Describe any problems you are currently experiencing with output reports” or “In your opinion, how helpful are the user manuals for the current system’s accounting application?”

When you write open-ended questions for a questionnaire, anticipate what kind of response you will get. For instance, if you ask a question such as, “How do you feel about the system?” the responses are apt to be too broad for accurate interpretation or comparison. Therefore, even when you write an open-ended question, it must be narrow enough to guide respondents to answer in a specific way. (Examples of open-ended questions can be found in Figure 4.11.)

FIGURE 4.11

Open-ended questions used for questionnaires.

53. What are the most frequent problems you experience with computer output?

A. _____

B. _____

C. _____

54. Of the problems you listed above, what is the single most troublesome?

55. Why?

Below are questions about yourself. Please fill in the blanks to the best of your ability.

67. How long have you worked for this company?
_____ Years and _____ Months

68. How long have you worked in the same industry?
_____ Years and _____ Months

69. In what other industries have you worked?

Open-ended questions can ask the respondent for lists ...

... or detailed responses ...

... or short answers.

FIGURE 4.12

Closed questions on questionnaires help ensure responses.

Answer questions 23 and 24 by checking the appropriate box.

23. Below are the six software packages currently available. Please check the software package(s) you personally use most frequently.

☐ Microsoft Excel	☐ Microsoft Access
☐ Microsoft PowerPoint	☐ Microsoft Project
☐ Oracle SCM	☒ Visible Analyst

24. "The sales figures are usually late."

☐ Agree	☒ Disagree
--------------------------------	--

Answer questions 25 and 26 by circling the appropriate number.

25. "When the sales figures are prepared by computer data services they are late."

Never	Rarely	Sometimes	Often	Always
1	2	3	4	5

Answer questions 45–48 by circling the appropriate response.

45. The division I am currently in is called

Investments
Operations
Marketing

46. My educational background can best be described as

High School
Some College
Bachelor's Degree
Master's Degree or Higher

My gender is

Male
Female

... or circle the answer itself.

... or circle a number ...

Closed questions may require the respondent to check a box ...

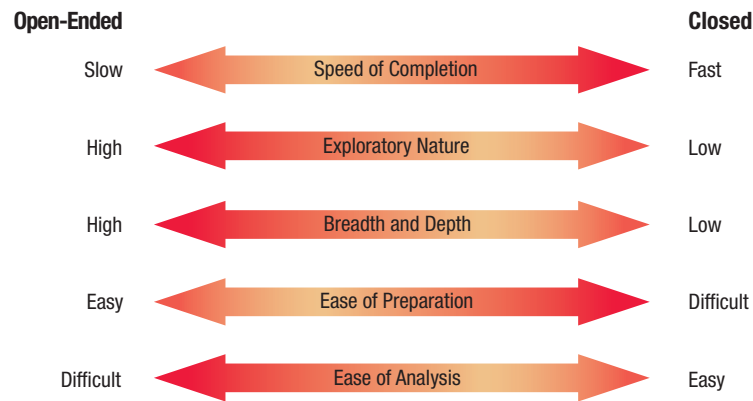
Open-ended questions are particularly well suited to situations in which you want to get at organization members' opinions about some aspect of the system, whether product or process. In such cases you will want to use open-ended questions when it is impossible to list effectively all the possible responses to the question.

CLOSED QUESTIONS Recall that closed questions (or statements) limit or close the response options available to the respondent. For example, in Figure 4.12 the statement in question 23 ("Below are the six software packages currently available. Please check the software package(s) you personally use most frequently.") is closed. Notice that respondents are not asked why the package is preferred, nor are they asked to select more than one, even if that is a more representative response.

A systems analyst should use closed questions when it is possible to effectively list all the possible responses to the question and when all the listed responses are mutually exclusive, so that choosing one precludes choosing any of the others.

FIGURE 4.13

Trade-offs between the use of open-ended and closed questions on questionnaires.



Use closed questions when you want to survey a large sample of people. The reason becomes obvious when you start imagining how the data you are collecting will look. If you use only open-ended questions for hundreds of people, correct analysis and interpretation of their responses becomes impossible without the aid of a computerized content analysis program or some type of text analytics software.

There are trade-offs involved in choosing either open-ended or closed questions for use on questionnaires. Figure 4.13 summarizes these trade-offs. Notice that responses to open-ended questions can help analysts gain rich, exploratory insights as well as breadth and depth on a topic. Although open-ended questions can be written easily, analyzing responses to them is difficult and time consuming.

When we refer to the writing of closed questions with either ordered or unordered answers, we often refer to the process as *scaling*. The use of scales in surveys is discussed in detail in a later section.

WORD CHOICE Just as with interviews, the language of questionnaires is an extremely important aspect of their effectiveness. Even if a systems analyst has a standard set of questions concerning systems development, it is wise to write them to reflect the business's own terminology.

Respondents appreciate the efforts of someone who bothers to write a questionnaire reflecting their own language usage. For instance, if the business uses the term *supervisors* instead of *managers* or *units* rather than *departments*, incorporating the preferred terms into the questionnaire helps respondents relate to the meaning of the questions. Responses will be easier to interpret accurately, and respondents will be more enthusiastic overall.

To check whether language used on the questionnaire is that of the respondents, try some sample questions on a pilot (test) group. Ask them to pay particular attention to the appropriateness of the wording and to change any words that do not ring true.

Here are some guidelines to use when choosing language for your questionnaire:

1. Use the language of respondents whenever possible. Keep wording simple.
2. Be specific rather than vague in your wording. Avoid overly specific questions as well.
3. Keep questions short.
4. Do not patronize respondents by talking down to them through low-level language choices.
5. Avoid bias in wording. Avoiding bias includes avoiding objectionable questions.
6. Target questions to the correct respondents (that is, those who are capable of responding). Don't assume too much knowledge.
7. Ensure that questions are technically accurate before including them.
8. Use software to check whether the reading level is appropriate for the respondents.

USING SCALES IN QUESTIONNAIRES Scaling is the process of assigning numbers or other symbols to an attribute or a characteristic for the purpose of measuring that attribute or characteristic. Scales are often arbitrary and may not be unique. For example, temperature is measured in a number of ways; the two most common are the Fahrenheit scale (where water freezes at 32 degrees and boils at 212 degrees) and the Celsius scale (where freezing occurs at 0 degrees and boiling at 100 degrees).



CONSULTING OPPORTUNITY 4.4

The Unbearable Questionnaire

“I’m going to go into a depression or at least a slump if someone doesn’t figure this out soon,” say Penny Stox, office manager for Carbon, Carbon, & Rippy, a large brokerage firm. Penny is sitting across a conference table from you and two of her most productive account executives, By Lowe & Sal Hy. You are all mulling over the responses to a questionnaire that has been distributed among the firm’s account executives, which is shown in Figure 4.C1.

“We need a crystal ball to understand these,” By and Sal call out together.

“Maybe it reflects some sort of optimistic cycle or something,” Penny says as she reads more of the responses. “Who designed this gem, anyway?”

“Rich Kleintz,” By and Sal call out in unison.

“Well, as you can see, it’s not telling us anything!” Penny exclaims.

Penny and her staff are dissatisfied with the responses they have received on the unbearable questionnaire, and they feel that the responses are unrealistic reflections of the amount of information account executives want. In a paragraph, state why these problems are occurring. On a separate sheet, change the scaling of the questions to avoid these problems.

Circle the appropriate number for each source of information described.

1. Industry Reports				
Less		About		More
1	2	the Same	4	5
		3		
2. Trend Analysis				
Less		About		More
1	2	the Same	4	5
		3		
3. Computer-Generated Graphs				
Less		About		More
1	2	the Same	4	5
		3		
4. Investment Advisory Services				
Less		About		More
1	2	the Same	4	5
		3		

We need to change this questionnaire.
-Penny

5. Point and Figure Charts				
Less		About		More
1	2	the Same	4	5
		3		
6. Computerized Portfolio Analysis				
Less		About		More
1	2	the Same	4	5
		3		
7. Hot Tips				
Less		About		More
1	2	the Same	4	5
		3		

FIGURE 4.C1

A questionnaire developed for the brokerage firm Carbon, Carbon, & Rippy by Rich Kleintz.

MEASUREMENT Systems analysts commonly use two different forms of measurement scales:

1. Nominal scales
2. Interval scales

Nominal scales are used to classify things. A question such as the following uses a nominal scale:

What type of software do you use the most?

- 1 = A word processor
- 2 = A spreadsheet
- 3 = A database
- 4 = An email program

Obviously, nominal scales are the weakest forms of measurement. Generally, all an analyst can do with them is obtain totals for each classification.

In interval scales, the intervals between each pair of numbers are equal. Due to this characteristic, mathematical operations can be performed on the questionnaire data, resulting in a more complete analysis. Examples of interval scales are the Fahrenheit and Celsius scales, which measure temperature.

By anchoring the scale on either end, the analyst may want to assume that the respondent perceives the intervals to be equal when they respond to this question:

How useful is the support given by the Technical Support Group?

<i>Not Useful at All</i>					<i>Extremely Useful</i>	
1	2	3	4	5		

If the systems analyst makes this assumption, more quantitative analysis is possible.

VALIDITY AND RELIABILITY There are two measures of performance in constructing scales: validity and reliability. A systems analyst should be aware of these concerns.

Validity is the degree to which a question measures what the analyst intends for it to measure. For example, if the purpose of the questionnaire is to determine whether the organization is ready for a major change in computer operations, do the questions measure that?

Reliability measures consistency. If a questionnaire was administered once and then again under the same conditions and if the same results were obtained both times, the instrument is said to have *external consistency*. If the questionnaire contains subparts and these parts have equivalent results, the instrument is said to have *internal consistency*. Both external and internal consistency are important.

CONSTRUCTING SCALES The actual construction of scales is a serious task. Careless construction of scales can result in the following problems:

1. Leniency
2. Central tendency
3. Halo effect

Leniency is a problem caused by respondents who are easy raters. A systems analyst can avoid the problem of leniency by moving the “average” category to the left (or right) of center.

Central tendency is a problem that occurs when respondents rate everything as average. The analyst can improve the scale (1) by making the differences smaller at the two ends, (2) by adjusting the strength of the descriptors, or (3) by creating a scale with more points.

The halo effect is a problem that arises when the impression formed in one question carries into the next question. For example, if you are rating an employee about whom you have a very favorable impression, you may give a high rating in every category or trait, regardless of whether it is a strong point of the employee’s. The solution is to place one trait and several employees on each page rather than one employee and several traits on a page.

Designing Questionnaires

Many of the same principles that are relevant to the design of forms for data input (as covered in Chapter 12) are important in designing questionnaires as well. Although the intent of a questionnaire is to gather information on attitudes, beliefs, behavior, and characteristics whose impact

may substantially alter users' work, respondents are not always motivated to respond. Remember that organization members on the whole tend to receive too many surveys, many of which are ill conceived and trivial.

A well-designed, relevant questionnaire can help overcome some of the resistance to responding. Here are some rules for designing a good questionnaire:

1. Allow ample white space.
2. Allow ample space to write or type in responses.
3. Make it easy for respondents to clearly mark their answers.
4. Be consistent in style.

When you design questionnaires for the Web, apply the same rules you use when designing paper questionnaires. Most software packages allow you to insert one of the commonly used data entry formats shown in Figure 4.14. Following the four guidelines should help you gain a better response rate to the questionnaire.

QUESTION ORDER There is no best way to order questions on a questionnaire. Once again, as you order questions, you must think about your objectives in using the questionnaire and then determine the function of each question in helping you to achieve your objectives. It is also important to see the questionnaire through the respondents' eyes. Some guidelines for ordering questions are:

1. Place questions that are important to respondents first.
2. Cluster items of similar content together.
3. Introduce less controversial questions first.

You want respondents to feel as unthreatened by and as interested in the questions being asked as possible, without getting overwrought about a particular issue.

Administering Questionnaires

The topic of administering questionnaires centers around two main questions. These are "Who in the organization should receive the questionnaire?" and "How should the questionnaire be administered?" Both questions are worth considering carefully before a questionnaire is sent. Doing so will help you better visualize and anticipate the data you are trying to collect with the questionnaire, and it also will help you plan ahead so that appropriate timing is observed in administering the questionnaire.

RESPONDENTS Deciding who will receive the questionnaire is handled in conjunction with the task of setting up objectives for its results. Sampling, which is covered in Chapter 5, helps a systems analyst determine what sort of representation is necessary and hence what kind of respondents should receive the questionnaire.

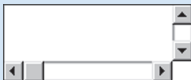
Name	Appearance	Purpose
One-line text box		Used to obtain a small amount of text and limit the answer to a few words
Scrolling text box		Used to obtain one or more paragraphs of text
Check box	☐	Used to obtain a yes-no answer (e.g., Do you wish to be included on the mailing list?)
Radio button	☐	Used to obtain a yes-no or true-false answer
Drop-down menu		Used to obtain more consistent results (Respondent is able to choose the appropriate answer from a predetermined list [e.g., a list of state abbreviations])
Push button		Most often used for an action (e.g., a respondent pushes a button marked "Submit" or "Clear")

FIGURE 4.14

When designing a Web survey, keep in mind that there are different ways to capture responses.



CONSULTING OPPORTUNITY 4.5

Order in the Courts

“I love my work,” Tennys says, beginning the interview with a volley. “It’s a lot like a game. I keep my eye on the ball and never look back,” he continues. Tennyson “Tennys” Courts is a manager for Global Health Spas, Inc., which has popular health and recreation spas worldwide.

“Now that I’ve finished my MBA, I feel like I’m on top of the world with Global,” Tennys says. “I think I can really help this outfit shape up with its computers and health spas.”

Tennys is attempting to help your systems group, which is developing a cloud-based system to be used by all 80 outlets. “Can

I bounce this off you?” he asks Terri Towell, a member of your team of systems analysts. “It’s a questionnaire I designed for distribution to all spa managers.”

Ever the good sport, Terri tells Tennys that she’d love to take a look at the form. But back in the office, Terri puts the ball in your court. Systematically critique Tennys’s technique, as depicted in Figure 4.C2, and explain to him point by point what it needs to be a matchless questionnaire with a winning form. Building on your critique, tell Tennys what he should do to rewrite the form as a Web survey instead.

QUESTIONNAIRE FOR ALL MANAGERS OF HEALTH SPAS

URGENTFILL OUT IMMEDIATELY AND RETURN PERSONALLY TO YOUR DIVISION MANAGER. YOUR NEXT PAYCHECK WILL BE WITHHELD UNTIL IT IS CONFIRMED THAT YOU HAVE TURNED THIS IN.

In 10 words or fewer, what complaints have you lodged about the current computer system in the last six months to a year?

Are there others who feel the same way in your outlet as you do? Who? List their names and positions.

1. 2.
3. 4.
5.
7.

What is the biggest problem you have when communicating your information requirements to headquarters? Describe it briefly.

How much computer downtime did you experience last year?
1 - 2 - 3 - 4 - 5 - 6 - 7 - 8 - 9 - 10 -

Is there any computer equipment you never use?

Description	Serial Number
Do you want it removed? Agree Neutral Disagree	
In your opinion, what's next as far as computers and Global Health Spas are concerned?	

Thanks for filling this out.

Terri
Please help
me improve this
form.
Tennys

FIGURE 4.C2

A questionnaire developed for managers of Global Health Spas by Tennys Courts.

Recipients are often chosen as representative because of their rank, length of service with the company, job duties, or special interest in the current or proposed system. Be sure to include enough respondents to allow for a reasonable sample in the event that some questionnaires are not returned or some response sheets are incorrectly completed and thus must be discarded.

METHODS OF ADMINISTERING A QUESTIONNAIRE A systems analyst has several options for administering a questionnaire, and the choice of administration method is often determined by the existing business situation. Options for administering the questionnaire include the following:

1. Convening all concerned respondents at one time.
2. Personally handing out blank questionnaires and taking back completed ones.



HYPERCASE EXPERIENCE 4.2

“You’ve probably noticed by now that not everyone enjoys filling out questionnaires at MRE. We seem to get more questionnaires than most other organizations. I think it’s because many of the employees, especially those from the old training unit, value the contributions of questionnaire data in our work with clients. When you examine the questionnaire that Mr. Evans distributed, you’ll probably want not only to look at the results but also to critique it from a methods standpoint. I always feel strongly that we can improve our internal performance so that eventually we can better serve our clients. The next time we construct a questionnaire, we want to be able to improve three things: the reliability of the data, the validity of the data, and the response rate we get.”

HYPERCASE Questions

1. What evidence of questionnaires have you found at MRE? Be specific about what you have found and where.
2. Critique the questionnaire that Mr. Evans circulated. What can be done to improve its reliability, validity, and response rate? Provide three practical suggestions.
3. Write a short questionnaire to follow up on some aspects of the merger between Management Systems and the Training Unit at MRE that are still puzzling you. Be sure to observe all the guidelines for good questionnaire design.
4. Redesign the questionnaire you wrote in Question 3 so that it can be used as a Web survey.

3. Allowing respondents to self-administer the questionnaire at work and drop it in a centrally located box.
4. Mailing questionnaires to employees at branch sites and supplying a deadline, instructions, and return postage.
5. Administering the questionnaire electronically either via email or on the Web.

Each of these five methods has advantages and disadvantages. Most commonly, respondents are allowed to self-administer a questionnaire. Response rates with this method are a little lower than with the other methods because people may forget about the form, lose it, or purposely ignore it. Self-administration, however, allows people to feel that their anonymity is ensured and may result in less guarded answers from some respondents. Both email and Web surveys fall into the category of self-administered questionnaires.

Administering a questionnaire electronically, either via email or posted on the Web, is one way to quickly reach current system users. Costs of duplication are minimized. In addition, responses can be made at the convenience of the respondent and then can be automatically collected and stored electronically. Some software permits respondents to begin answering a survey, save their answers, and return to it for completion if they are interrupted. Reminders to respondents can be easily and inexpensively sent via email, as can notifications to the analyst about when the respondent has opened the email. Some software now turns email data into data tables for use in spreadsheet or statistical analysis software. One of the popular services for creating and administering online surveys is a private company called SurveyMonkey (see www.surveymonkey.com). It is a cloud-based Software as a Service (SaaS) company that acquired an email marketing services company called MailChimp to add to its product offerings. MailChimp allows organizations to send email newsletters and design, send, and track email campaigns.

Research shows that respondents are willing to answer questions about highly sensitive matters via the Internet. Thus, questions that may be difficult to pose in person regarding systems problems may be acceptable to ask on a Web survey.

Summary

This chapter covers three of the key interactive methods for information gathering that the systems analyst can use: interviewing, joint application design, and construction of questionnaires. During the process of interviewing analysts, a systems analyst should listen for HCI concerns related to ergonomics, aesthetics, usability, and usefulness, as well as goals, feelings, opinions, and informal procedures in interviews with

organization decision makers. Interviews are planned question-and-answer dialogues between two people. An analyst can use an interview to develop a relationship with a client, to observe the workplace, and to collect data. Interviews should preferably be conducted in person.

The five steps to take in planning an interview are to read background material, establish interviewing objectives, decide who to interview, prepare the interviewee, and decide on question types and structure.

Questions are of two basic types: open-ended or closed. Open-ended questions leave open all response options for the interviewee. Closed questions limit the possible options for response. Probes or follow-up questions can be either open-ended or closed, but either way, they ask the respondent for a more detailed reply.

Interviews can be structured in three basic ways: pyramid, funnel, or diamond. Pyramid structures begin with detailed, closed questions and broaden to more generalized questions. Funnel structures begin with open-ended, general questions and then funnel down to more specific, closed questions. Diamond-shaped structures combine the strengths of the other two structures, but they take longer to conduct. Trade-offs are involved when deciding how structured to make interview questions and question sequences.

To cut both the time and cost of personal interviews, analysts may want to consider joint application design instead. Using JAD, analysts can both analyze human information requirements and design a user interface with users in a group setting. Careful assessment of the particular organization culture will help an analyst judge whether JAD is suitable.

By using questionnaires (surveys), systems analysts can gather data on HCI concerns, attitudes, beliefs, behavior, and characteristics from key people in the organization. Surveys are useful if people in the organization are widely dispersed, if many people are involved with the systems project, if exploratory work is necessary before recommending alternatives, and if there is a need for problem sensing before interviews are conducted.

Once objectives for the survey are set, the analyst can begin writing either open-ended or closed questions. Ideally, the questions should be simple, specific, short, free of bias, not patronizing, technically accurate, addressed to those who are knowledgeable, and written at an appropriate reading level. The systems analyst may want to use scales either to measure the attitudes or characteristics of respondents or to have respondents act as judges for the subject of the questionnaire. Scaling is the process of assigning numbers or other symbols to an attribute or characteristic.

Consistent control of the questionnaire format and style can result in a better response rate. Web surveys can be designed to encourage consistent responses. In addition, the meaningful ordering and clustering of questions is important for helping respondents understand the questionnaire. Surveys can be administered in a variety of ways, including electronically via email or the Web, or with the analyst present in a group of users. Software or SaaS is available to automatically create, distribute, and then tabulate email or Web responses.

Systems analysts can use storytelling as a complement to other information-gathering methods. You need to involve users by reacting to stories, matching one story to another by recounting it to other participants, and even collaborating with the participant to reframe and understand organization stories. It is a useful way to understand problems associated with the use of technology, systems development, systems adoption, and systems design. The systems analyst needs to listen to the story as a whole to truly understand its content and purpose.

Keywords and Phrases

bipolar closed questions
central tendency
closed questions
diamond-shaped structure
funnel structure
halo effect
human-computer interaction (HCI)
informal procedures
interval scale
interviewee feelings
interviewee goals
interviewee opinions

joint application design (JAD)
leniency
nominal scale
open-ended questions
probes
pyramid structure
questionnaire
reliability
stories
survey respondents
validity

Review Questions

1. Why is it important to seek opinions during interviews?
2. Describe a situation in which JAD would be ideal.
3. When should an interviewer compose a written interview report and why?
4. When are open-ended questions appropriate for use in interviewing?
5. Define what is meant by closed interview questions. List six benefits and four drawbacks of using them.
6. When are closed questions appropriate for use in interviewing?
7. What is a probing question? What is the purpose of using a probing question in interviews?
8. Define what is meant by *pyramid structure*. When is it useful to employ this structure in interviews?
9. Define what is meant by *funnel structure*. When is it useful to employ this structure in interviews?
10. Define what is meant by *diamond-shaped structure*. When is it useful to employ this structure in interviews?
11. Compose an open-ended question for Dr. DiVito, a university professor who is eligible for retirement in three weeks.
12. Discuss the importance of user stories for systems analysts.
13. Define *joint application design (JAD)*.
14. List the situations that warrant use of JAD in place of personal organization interviews.
15. Identify the various participants involved in a JAD session.
16. What does IBM recommend for the structured approach of JAD? Where would it be less useful?
17. What kinds of information is a systems analyst seeking through the use of questionnaires or surveys?
18. List four situations in which the use of questionnaires is appropriate.
19. In what sense does choosing either open-ended or closed-ended questions involve a trade off?
20. List two reasons a systems analyst would use a closed question on a questionnaire.
21. List two reasons a systems analyst would use an open-ended question on a questionnaire.
22. Explain how the language of questionnaires is an important aspect of their effectiveness.
23. Define what is meant by *scaling*.
24. What are two kinds of information or scales that systems analysts most commonly use?
25. What are nominal scales used for?
26. Give an example of an interval scale.
27. When should an analyst use an interval scale?
28. Define *reliability* as it refers to the construction of scales.
29. Define *validity* as it refers to the construction of scales.
30. List three problems that can occur because of careless construction of scales.
31. What four actions can be taken to ensure that a questionnaire's format is conducive to a good response rate?
32. What are the possible benefits of administering questionnaires electronically?
33. Why should questions on similar topics be clustered together?
34. What is an appropriate placement for controversial questions?
35. List five methods for administering a questionnaire.
36. What considerations are necessary when questionnaires are Web-based?

Problems

1. As part of your systems analysis project to update the automated accounting functions for Xanadu Corporation, a manufacturer of digital cameras, you will interview Leo Blum, the chief accountant. Write four to six interview objectives covering his use of information sources, information formats, decision-making frequency, desired qualities of information, and decision-making style.
 - a. In a paragraph, write down how you will approach Leo to set up an interview.
 - b. State which structure you will choose for this interview. Why?
 - c. Leo has four subordinates who also use the system. Would you interview them also? Why or why not?
 - d. Would you also try to interview customers (visitors to the website)? Are there better ways to get the opinions of customers? Why or why not?
 - e. Write three open-ended questions that you will email to Leo prior to your interview. Write a sentence explaining why it is preferable to conduct an interview in person rather than via email.
 - f. Describe how you would ask Leo to tell a relevant story. What seven elements should you listen for?
2. Here are five questions written by one of your systems analysis team members. Her interviewee is the local manager of LOWCO, an outlet of a national discount chain, who has asked you to work on a management information system to provide inventory information. Review these questions for your team member.

1. When was the last time you thought seriously about your decision-making process?
2. Who are the trouble makers in your store—I mean the ones who will show the most resistance to changes in the system that I have proposed?
3. Are there any decisions you need more information about before you can make them?
4. You don't have any major problems with the current inventory control system, do you?
5. Tell me a little about the output you'd like to see.
 - a. Rewrite each question so it is more effective in eliciting information.
 - b. Order your questions in either a pyramid, funnel, or diamond-shaped structure and label the questions with the name of the structure you used.
 - c. What guidelines can you give your team member for improving her interviewing questions in the future? Make a list of them.
3. Ever since you walked through the door, your interviewee, Max Hugo, has been shuffling papers, looking at his watch, and drumming on his desk with his fingers. Based on what you know about interviews, you guess that Max is nervous because of the other work he needs to do. In a paragraph, describe how you would deal with this situation so that the interview can be accomplished with Max's full attention. (Max cannot reschedule the interview for a different day.)
4. Write a series of six *closed* questions that cover the subject of decision-making style for the manager described in Problem 2.
5. Write a series of six *open-ended* questions that cover the subject of decision-making style for the manager described in Problem 2.
6. Examine the list of possible interview scenarios presented below:
 1. Reasons for absenteeism from work
 2. Why job candidates wish to work at a firm
 3. Customers' interview on the addition of a "features" section on a website
 4. Employee reactions to a change in management
 5. An interview by executives of a consultant who is presenting his findings
 6. Local vendors' opinions on foreign partnerships in organic foods marketing
 7. Suppliers' opinions on ways to reduce costs
 8. Home-owner attitudes toward changes in a loan process
 9. Ethical dilemmas employees face at work
 - a. Explain which interview structure would be appropriate for each scenario.
 - b. Prepare a list of interview questions for at least two of the scenarios depicted above. Is it possible to reorder the questions and change the interview structure? Justify your answer.
7. The following is a story narrated to a systems analyst involved in developing an information system for selecting and ordering a book for a library. "A library employed a branch librarian and five librarians. Each month, the librarians placed a check mark against a list of forthcoming books received for pre-order. The branch librarian would order books with at least three check marks after exercising his own judgment. However, the librarians believed that the head librarian wasn't using his judgment at all. Moreover, as there was only one librarian who handled children's books, they were rarely ordered. They proved their point by placing a check mark against a highly priced book that was not relevant to the library, yet the book was ordered and received."
 - a. What is the type of user story narrated above?
 - b. Discuss how a systems analyst can use this story for information system design and development.
 - c. Should a systems analyst complement this story with other information gathering methods? Why?
8. Mike is a newly hired systems analyst with your group. He has always believed that JAD sessions are a huge waste of professional time as they need lots of preparation and commitment. You will be working on a systems project for Ken University that will enroll international students through its distance programs, and you want to use JAD sessions to elicit opinions from management and students about the current and proposed systems.
 - a. Based on what you know about the requirements of the project, list three reasons why Mike should use JAD for this study.
 - b. Given your arguments, Mike agrees to use JAD but strongly urges that it be made as structured as possible. In a paragraph, illustrate the approach you would follow to implement JAD. How would you structure it?
9. Experiencing increased employee turnover after implementing a new information system, Global Marketing Inc. decided to distribute a standard questionnaire to gauge reasons for employee satisfaction or dissatisfaction. However, employees said that many of the points covered by the questionnaire were either irrelevant or insufficient. Additionally, some employees were reluctant to share their opinions. Consider this situation as a systems analyst and answer the following:
 - a. Do you think a custom questionnaire would be an improvement in this situation? How?
 - b. How would soliciting user stories before designing the questionnaire make it more relevant?
 - c. What would be the best way to administer the questionnaire? Would electronic administration offer greater benefits?

10. A sample question from the draft of the Pohattan Power Company questionnaire reads:
- I have been with the company:
- 20–upwards years
10–15 years upwards
5–10 years upwards
less than a year
- Check one that most applies.
- What kind of a scale is the question's author using?
 - What errors have been made in the construction of the question, and what might be the possible responses?
 - Rewrite the question to achieve clearer results.
 - Where should the question you've written appear on the questionnaire?
11. Also included on the PPC questionnaire is this question:
- When residential customers call, I always direct them to our website to get an answer.

<i>Sometimes</i>	<i>Never</i>	<i>Always</i>	<i>Usually</i>
1	2	3	4

- What type of scale is this intended to be?
 - Rewrite the question and possible responses to achieve better results.
12. Figure 4.EX1 is a questionnaire designed by an employee of Green Toe Textiles, which specializes in manufacturing men's socks. Di Wooly wrote the questionnaire because, as the office manager at headquarters in Juniper, Tennessee, she is concerned with the proposed purchase and implementation of a new computer system.
- Provide a one-sentence critique for each question given.
 - In a paragraph, critique the layout and style in terms of white space used, room for responses, ease of responding, and so on.
13. Based on what you surmise Ms. Wooly is trying to get through the questionnaire, rewrite and reorder the questions (use both open-ended and closed questions) so that they follow good practice and result in useful information for the systems analysts. Indicate next to each question that you write whether it is open-ended or closed and write a sentence indicating why you have written the question this way.
14. Redesign the questionnaire you created for Ms. Wooly in Problem 13 for use on email. Write a paragraph that says what changes were necessary to accommodate email users.
15. Redesign the questionnaire you created for Ms. Wooly in Problem 13 as a Web survey. Write a paragraph that says what changes were necessary to accommodate Web users.

Hi! All Employees

What's new? According to the grapevine, I hear we're in for a new computer. Here are some questions for you to think about.

- How long have you used the old computer? _____
- How often does it go down? _____
- Who repairs it for you? _____
- When was the last time you suggested a new improvement to the computer system and it was put into use? What was it? _____
- When was the last time you suggested a new improvement to the computer system and nobody used it? What was it? _____
- Do you use a VDT or printer or both? _____
- How fast do you type? _____
- How many people need to access the database regularly at your branch? Is there anyone not using the computer now who would like to? _____

FIGURE 4.EX1

A questionnaire developed by Di Wooly.

Group Projects

1. With your group members, role-play a series of interviews with various system users at Maverick Transport. Each member of your group should choose one of the following roles: company president, information technology director, dispatcher, customer service agent, or truck driver. Those group members playing roles of Maverick Transport employees should attempt to briefly describe their job responsibilities, goals, and informational needs.

Remaining group members should play the roles of systems analysts and devise interview questions for each employee. If there are enough people in your group, each analyst may be assigned to interview a different employee. Those playing the roles of systems analysts should work together to develop common questions that they will ask, as well as questions tailored to each individual employee. Be sure to include open-ended, closed, and probing questions in your interviews.

Maverick Transport is attempting to change from outdated and unreliable technology to more state-of-the-art, dependable technology. The company is seeking to move from PCs connected to a mainframe because it wants to use tablets to improve worker mobility, and it is also interested in investigating a cloud-based system for tracking freight and drivers. In addition, the company is interested in pursuing ways to cut down on the immense storage requirements and difficult access of the troublesome handwritten, multipart forms that accompany each shipment.

2. Conduct all five interviews in a role-playing exercise. If there are more than 10 people in your group, permit two or more analysts to ask questions. Remember to devote part of the interview to asking for and listening to stories from each interviewee.
3. With your group, write a plan for a JAD session that takes the place of personal interviews. Include relevant participants, suggested setting, and so on.
4. Using the interview data you gained from the group exercise on Maverick Transport in Group Project 1, meet with your group to brainstorm the design of a questionnaire for the hundreds of truck drivers that Maverick Transport employs. Recall that Maverick is interested in implementing a cloud-based system for tracking freight and drivers. There are other systems that may affect the drivers as well. As your group constructs the questionnaire, consider the drivers' likely level of education and any time constraints the drivers are under for completing such a form.
5. Using the interview data you gained from the group exercise on Maverick Transport in Group Project 1, your group should meet to design an email or Web questionnaire for surveying the company's 20 programmers (15 of whom have been hired in the past year) about their skills, ideas for new or enhanced systems, and so on. Investigate the Web survey options available at SurveyMonkey.com. As your group constructs the computer programmer survey, consider what you have learned about users in the other interviews as well as what vision the director of information technology has for the company.
6. Divide your group into three distinct teams considering the case of a large manufacturer and online seller of sports clothing and health accessories called Ganymede Fitness. They manufacture and sell clothing including shirts, shorts, pants, headbands, and some fitness technology products. They are privately held, global company headquartered in Seattle, Washington, in the U.S., which has a corporate slogan stating, "Helping the world get fit, happy, and healthy, one person at a time."
 - a. Assign team members so that there are an equal number of team members in each of the following three groups:

Team 1 Systems Analysts. This team should take the position of systems analysts who have been asked to gather feelings of two groups—Team 2, the consumers of Ganymede Fitness products and their website content, and asking Team 3, the Ganymede Fitness ecommerce managers about the usefulness of recommendation systems and ratings systems for online shopping. Eventually, the analysts will be tasked with designing recommendation systems and rating systems for Ganymede Fitness.

Team 2 Ganymede Fitness product users and website users. This team is made up of avid users of Ganymede Fitness products (which are only available on their website), many of who have been with them since their opening in 2010. They have eagerly adopted all of the products Ganymede Fitness offered, most recently their fitness technology products.

Team 3 Ganymede Fitness Ecommerce managers. This team is composed of ecommerce managers who work for Ganymede Fitness and are charged with keeping the ecommerce portion of the website up to date. They would like to know more about their customers so that the website becomes more personalized with recommendations for purchases and ratings of their products. They would also like to find out from the systems analysts what the possibilities are for improving their website.

- b. Each team should meet virtually or face to face and reaffirm their purpose as a team. During their initial meeting, Team 1 should prepare a series of interview questions for both Teams 2 and 3. Team 1 of systems analysts should prepare a list of six open-ended questions for the users and a list of six open-ended questions for the ecommerce managers. Team 2 of product and website

- users should discuss and create a common list of sports clothing and health accessories they have purchased or would consider purchasing online, and what their favorite features of each product are. Team 3 of ecommerce managers should write six interview questions for Team 1 about what kind of support for online purchases, including ratings and recommendations, they can provide. Team 3 should also prepare six questions for Team 2 about their website preferences.
- c. Once each team has completed their assignment, they should meet face to face or in a monitored chat room (using whatever technology is available to the class) to role-play the tasks and perspectives outlined for each team in part a. Team 1 should take the lead and begin interviewing members of Team 2. Team 3 should join in asking any questions that have not yet been asked of Team 2, and also following up on the initial questions with probing questions. Team 1 should move to the next phase, and ask their prepared questions of Team 3 about how useful ratings and recommendations would be to the Ganymede Fitness website. Assigning a scribe or note taker for each team to write notes about answers will help move the process along for the last step.
 - d. Each of the three teams should meet separately (face to face or online) as a team to discuss what they learned in response to their questions. Each team should write a brief one-page report that includes: date, time of day, length of interview, and participants in the interview. The one-page report should include a summary of key information learned in the interview, and also include a list of any points that were raised but that still need to be clarified.
7. To extend exercise six above, teams working on the Ganymede Fitness case can share their written reports with the other two teams (via posting on the course website, or via email) to compare where they are in terms of understanding the usefulness of improving the ecommerce website for Ganymede Fitness with product ratings and recommendations. Teams can make a brief oral presentation of what they learned in the interview process including (1) any suggestions for new material for the website and (2) any follow up questions they would like to ask and which team members they would like to ask additional questions.
 8. Each of the three teams working on the Ganymede Fitness case above should design an online survey based on their experience with conducting interviews. The online survey should be developed for a specific audience, either Team 2 of Ganymede Fitness product and website users or Team 3 of Ganymede Fitness ecommerce managers. The online survey should pose both open ended and closed questions about the desirability of adding product rankings and recommendations to the Ganymede Fitness website.

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